

AI

Chapter-8- Network

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Q) What do you mean by Network and Computer Network?

A) Network means group of any entity such as group of people, group of schools etc.

• This group ~~which~~ will always consist two or more entities

• Computer network is a group of connected computers, computing devices. This group of connected computers are able to share data and other resources within the group through a connected network (wired or wireless)

Q) Enlist any four advantages of computer network.

* All IOT devices

- Most of the AI enabled devices works with the Internet so that the device can function well. Network helps AI devices to collect information through sensors and respond according to it.

* Hardware Sharing

- Computer network helps in sharing multiple hardware components connected within same network.

* Low cost requirement for data transfer

- Computer network reduces the cost of data transfer from one place to another with an established network.

SMTP - Simple Mail Transfer Protocol.

IMAP - Internet message access protocol

POP3 - Post Office Protocol version 3

* Secured data transfer

Computer network uses high-
speeded protocols to send
files from one place to
another

GEO.

Do complete the last page
as it's a homework!!

at 12/25

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Cold weather or Winter season

- It begins from a November and continues upto end of February.
- Days are short and nights become longer.
- Temperature decreases from South to north with southern India having absence of winter season while northern India has a prominent winter season.
- This season is characterized by bright and clear sky with warm ~~and~~ day temp. and cold nights.
- Northern India experiences fog in the morning hours.

- During this period east coasts get cyclonic rains brought by N-E Trade winds.

- The Hill Stations of Himalayas gets snowfall during this time.

- The average temp. of the Indian ~~sub~~ is around $15-18^{\circ}\text{C}$

Hot weather/ Summer season

- The Hot weather of the summer season. It begins from March and continues upto May.

- Temperature increases from south to North with many regions of Northern India experience a temperature above 40°C .

- This season is marked by clear sky and scorching sunshine.
- The Northern Plains experience loo winds that blow during afternoon hours making the afternoon heat unbearable.
- In the month of April and May pre monsoonal showers called the Mango showers of the west coast and Kal Baisakhi of West Bengal, Orissa & Chhatisgarh occurs in the afternoon hours.
- The Kal Baisakhi are devastating thunderstorms that causes huge destructions but brings some relief from the scorching heat.

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Monsoon season

Advancing
Monsoon

Retreating
monsoon

This is homework!!

ENG.

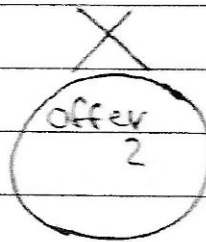
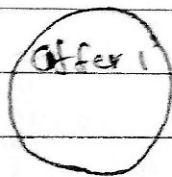
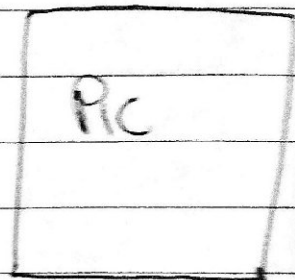
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PHY.

Shadow & Reflection

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Reflection

Bouncing back of light on the same medium by a light source.

This process is known as Reflection.

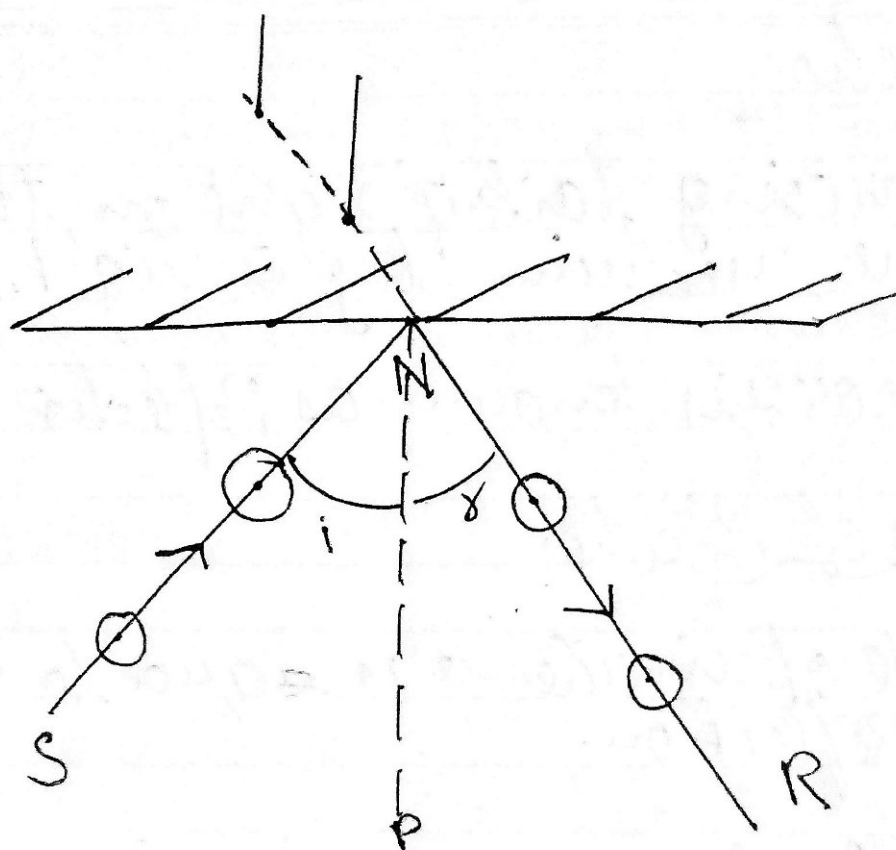
Laws of Reflection

Angle of incidence is equal to Angle of Reflection.

Incident ray, normal and reflected ray lie at the same point of incidence of same plane.

Incident ray - When a ray is striking on a plane mirror is known as Incident ray.

Reflected ray - After striking on the plane mirror if ray returns back to the same medium is known as reflected ray.



SN - Incident Ray

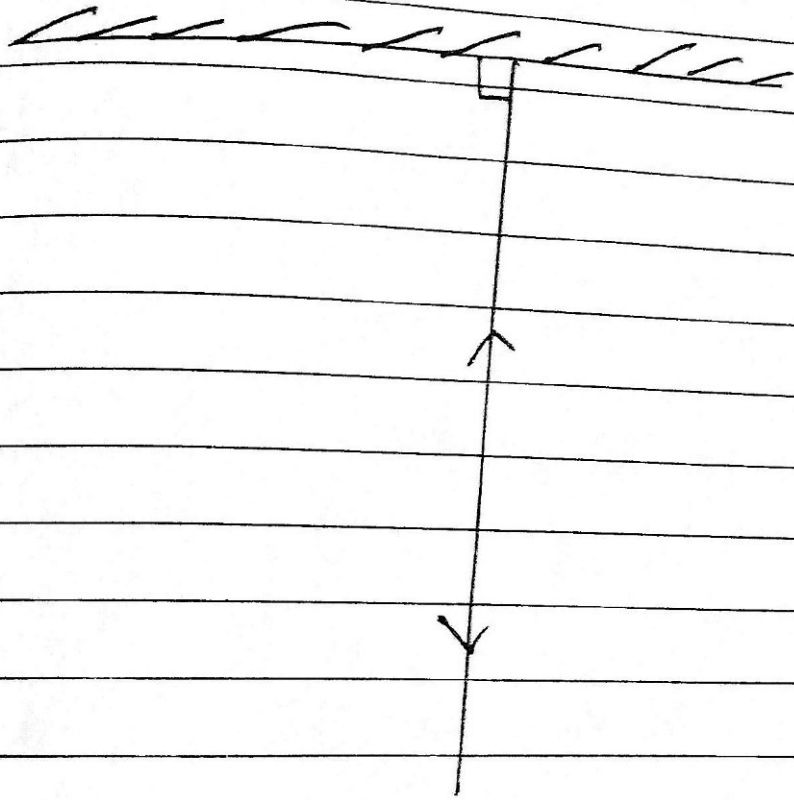
NR - Reflected Ray

NP $\rightarrow$ Normal

$i \rightarrow$ Angle of incidence

$r \rightarrow$ Angle of Reflection

Normal. An Imaginary line $\perp$ to
plane mirror is normal
NP $\rightarrow$ normal.



MATH

(iii) Area of an equilateral triangle with side a is

$$A = \left(\frac{\sqrt{3}}{4} a^2 \right) \text{ Sq. unit}$$

Heron's formula:

Let a, b, c be the sides of $\triangle ABC$
 Then $s = \frac{1}{2}(a + b + c)$

$$\text{Area of } \triangle ABC = \sqrt{s(s-a)(s-b)(s-c)} \text{ sq. unit}$$

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(Q) Find the area of a $\triangle$ in which
 base = 25cm and height = 14cm.

$$b = 25\text{cm} \quad h = 14\text{cm}$$

$$\begin{aligned} \text{Area of a triangle} &= \left(\frac{1}{2} \times b \times h \right) \\ &= \left(\frac{1}{2} \times 25 \times 14 \right) \text{cm}^2 \\ &= 175\text{cm}^2 \end{aligned}$$

89 The base of a Δ field is 3 times its height. If the cost of cultivating a field at ₹ 2800/hectare is ₹ 37800, find its base and height.

$$\text{Total cost} = ₹ 37800$$

$$\text{Rate} = ₹ 2800/\text{hectare}$$

$$\text{Area of the field} = \left(\frac{\text{Total cost}}{\text{Rate}} \right) \text{ hectares}$$

$$= \left(\frac{37800}{2800} \right) \text{ hectares}$$

$$= \left(\frac{27}{2} \right) \text{ hectares}$$

$$= \left(\frac{27}{2} \times 10000 \right) \text{ m}^2$$

$$= \boxed{135000 \text{ m}^2}$$

Let the height be u m then it's
base = $3u$ m

$$\begin{aligned}\text{Area of the field} &= \left(\frac{1}{2} \times 3u \times u\right) \text{ m}^2 \\ &= \left(\frac{3u^2}{2}\right) \text{ m}^2\end{aligned}$$

$$\therefore \frac{3u^2}{2} = 135000$$

$$\Rightarrow u^2 = \left(\frac{45000}{135000} \times \frac{2}{3}\right)$$

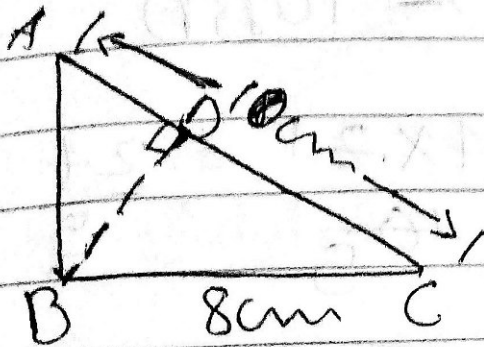
$$\Rightarrow u^2 = 90,000$$

$$\Rightarrow u = \sqrt{90,000}$$

$$\therefore u = \boxed{300}$$

So height of the field = 300 m
base = (3×300) m = 900 m

Q) Find the area of ΔABC in which $\angle B = 90^\circ$, $BC = 8\text{cm}$ and $AC = 10\text{cm}$ if $BO \perp AC$ then find BD .



gm rt. ΔABC

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow (10)^2 = (AB)^2 + (8)^2$$

$$\Rightarrow AB^2 = 100 - 64$$

$$\Rightarrow AB^2 = 36$$

$$\Rightarrow AB = \sqrt{36}$$

$$\therefore AB = \boxed{6\text{cm}}$$

$$\text{Area of } \Delta ABC = \left(\frac{1}{2} \times 8 \times 6 \right) \text{cm}^2$$

$$= \boxed{24\text{cm}^2}$$

Now area of $\triangle ABC = \frac{1}{2} \times AC \times BD$

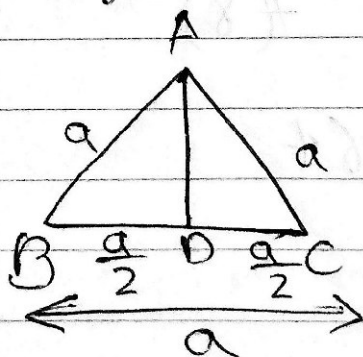
$$\Rightarrow 24 = \frac{1}{2} \times 10 \times BD$$

$$\Rightarrow 24 \times 2 = 10 BD$$

$$\Rightarrow BD = \frac{24 \times 2}{10} = \frac{24}{5}$$

$$= \boxed{4.8 \text{ cm}}$$

Finding the area of an equilateral $\triangle$.



In rt. $\triangle AOB$;

$$AB^2 = BD^2 + AD^2$$

$$\Rightarrow a^2 = \left(\frac{a}{2}\right)^2 + AD^2$$

$$\Rightarrow a^2 = \frac{a^2}{4} + AD^2$$

$$\Rightarrow AD^2 = a^2 - \frac{a^2}{4}$$

$$\Rightarrow AD^2 = \frac{4a^2 - a^2}{4} = \frac{3a^2}{4}$$

$$\Rightarrow AD = \frac{\sqrt{3a^2}}{2}$$

$$\therefore AD = \frac{\sqrt{3} a}{2}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times a \times \frac{\sqrt{3} a}{2}$$

$$= \boxed{\frac{\sqrt{3} a^2}{4}}$$

- Q) Find the area of an equilateral $\triangle$ each of whose side is 10 cm.
[Take $\sqrt{3} = 1.73$]

$$\text{Area of an eq. } \triangle = \frac{\sqrt{3} \times a^2}{4}$$

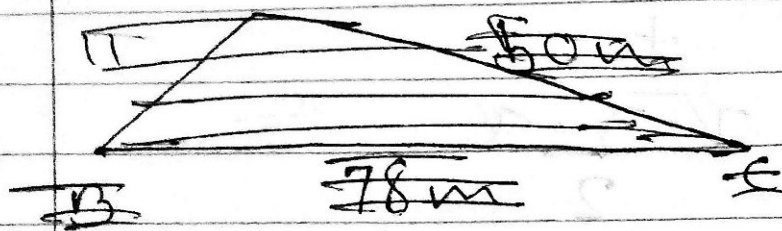
$$= \frac{\sqrt{3} \times 10 \times 10 \text{ cm}^2}{4}$$

$$= \frac{100 \sqrt{3} \text{ cm}^2}{4}$$

$$= \frac{100 \times 1.73 \text{ cm}^2}{4} = \frac{100 \times 1.73}{4 \times 100}$$

$$= \frac{173}{4} \text{ cm}^2 = 43.25 \text{ cm}^2$$

Q) Find the area of a Δ field whose sides are 78m, 50m, & 112m.



$$\left[\frac{50 \times 78}{2} \right] =$$

$$1950 \text{ m}^2 = \Delta \text{ area}$$

$$1950 \times 2 = 3900$$

$$3900 \div 2 = 1950$$